# Data-Driven Hyperparameter Optimization for Accurate Sparse Deconvolution of Spatial Transcriptomic Data

Habiba Hammouda
*Biomedical Engineering Department,*
*Cairo University, Giza, Egypt*
habiba.hammouda03@eng-st.cu.edu.eg

## ABSTRACT

**Motivation:** Spatial transcriptomics platforms such as 10X Visium measure gene expression in tissue while preserving spatial context, but each spot captures multiple cells, producing mixed transcriptional profiles. Computational deconvolution methods infer the underlying cell-type composition, but their performance depends critically on hyperparameter selection. WISpR (Weight-Induced Sparse Regression) employs cross-validated GridSearch to optimize threshold ($\tau$) and penalty ($\lambda$) parameters independently for each spot. Whether this computationally expensive strategy is essential, or whether manually selected parameters suffice, has not been rigorously tested.

**Results: GridSearch-optimized WISpR was compared against four manually selected hyperparameter configurations on mouse hippocampus Visium data (2,110 spots, 26 reference cell types). GridSearch produced biologically realistic sparsity (mean 4.22 cell types per spot) and low reconstruction error (median RMSE = 0.280), with predicted proportions strongly correlated to canonical marker genes (oligodendrocytes r = 0.95, immune r = 0.81, vascular r = 0.79–0.80; all $p < 10^{-230}$). All four manual configurations performed substantially worse: 41.6%–95.7% higher RMSE, 3–6× more cell types per spot than biologically plausible, and Cohen's d ranging from −0.77 to −2.76 (all Wilcoxon $p < 10^{-100}$). Cliff's delta values near −1.0 indicate the GridSearch baseline outperforms manual selection on essentially every spot. These results demonstrate that uniform hyperparameters cannot accommodate spatially heterogeneous tissue biology and that data-driven, spot-specific optimization is required for biologically meaningful deconvolution.**

**Availability: Code and processed data are available at** https://github.com/HabibaHammouda/wispr.git

**Keywords:** *spatial transcriptomics; deconvolution; WISpR; hyperparameter optimization; sparse regression; Visium; hippocampus*

## 1 INTRODUCTION

Spatial transcriptomics technologies measure gene expression while preserving the spatial organization of tissues, providing a powerful complement to single-cell RNA sequencing (scRNA-seq) [1–3]. Platforms such as 10X Genomics Visium, Slide-seq, and MERFISH have enabled the systematic mapping of cellular architecture in development, disease, and tissue regeneration [4]. However, most spatial platforms operate at resolutions that capture multiple cells per measurement: each 55 μm Visium spot, for example, typically contains 1–10 cells whose transcriptomes are sequenced together. The resulting mixed profiles obscure individual cell-type contributions, motivating the development of computational deconvolution methods that infer cell-type composition from mixed spatial measurements [5].

A growing family of methods now addresses this problem. Early approaches extended bulk RNA-seq deconvolution, such as dampened weighted least squares (DWLS) [6], which estimates proportions by minimizing weighted residuals against a reference signature matrix. More recent methods incorporate platform-aware statistical models: RCTD [7] uses a Poisson log-normal mixture; SPOTlight [8] applies seeded non-negative matrix factorization; Stereoscope [9] employs a negative binomial generative model; and Cell2location [10] uses a Bayesian hierarchical framework with explicit spatial priors. Although these methods differ in their statistical formulations, they share a practical requirement: each depends on hyperparameters that govern model complexity, sparsity, or regularization.

WISpR (Weight-Induced Sparse Regression) is a recent sparse deconvolution algorithm that explicitly enforces biological sparsity—the constraint that each spatial measurement should contain only a small subset of reference cell types [11]. WISpR uses an iterative thresholding procedure inspired by the SINDy framework [12], progressively pruning weakly supported cell types until only the most strongly supported remain. This is biologically appropriate for Visium data, where physical spot size dictates that only 3–5 cell types should typically be detectable per spot; predictions of 10 or more cell types per spot suggest overfitting rather than biology.

WISpR's behavior is governed by two hyperparameters: a thresholding parameter $\tau$ that controls how aggressively weak predictions are eliminated, and a penalty parameter $\lambda$ that controls L2 regularization. The published implementation optimizes both parameters independently for each spot via GridSearchCV with cross-validation. This is computationally expensive, and in practice some users have substituted manually chosen, uniform hyperparameters based on biological intuition. Whether this substitution is justifiable—or whether spot-specific optimization is genuinely necessary—has not been systematically evaluated.

Resolving this question matters beyond WISpR. Most spatial deconvolution methods rely on hyperparameters whose optimal values are tissue-, platform-, and dataset-dependent. If data-driven optimization is critical, the field should standardize on cross-validated tuning; if manual choices suffice, considerable computation can be avoided. In this work, GridSearch-optimized WISpR is compared against four manually selected hyperparameter configurations spanning plausible choices (loose threshold, strict threshold, weak penalty, strong penalty) on a mouse hippocampus Visium dataset paired with a matched scRNA-seq reference atlas. Reconstruction accuracy (RMSE), biological sparsity, marker-gene validation, spatial coherence, and statistical significance were assessed across 2,110 spots.

Data-driven optimization is found to be not merely beneficial but essential. All manual configurations produced 41.6%–95.7% higher RMSE than the GridSearch baseline, with effect sizes far exceeding conventional thresholds for large effects (Cohen's d up to −2.76). Even the best-performing manual configuration predicted nearly three times the biologically realistic number of cell types per spot. These results establish that uniform hyperparameters cannot accommodate the spatial heterogeneity of real tissues and that adaptive, spot-specific optimization should be considered a standard requirement for WISpR and likely for related sparse deconvolution methods.

## 2 MATERIALS AND METHODS

### *2.1 Datasets*

Mouse brain hippocampus tissue profiled with the 10X Genomics Visium platform was analyzed. The dataset comprised 2,110 spots subsampled from the original ~2,700 spots to accommodate memory constraints on an 8 GB workstation, with a mean of 31,012 UMIs and 6,036 detected genes per spot. Spot coordinates were preserved from the 10X tissue position file; each spot is 55 μm in diameter and captures approximately 1–10 cells.

For the single-cell reference, the L1 hippocampus dataset (l1_hippocampus.loom) from the Linnarsson Laboratory Mouse Brain Atlas [13] was used. The original dataset contains 29,120 cells across 138 hierarchically defined clusters. Following Erdogan and Eroglu [11], only clusters containing 25–250 cells were retained to ensure stable mean expression estimates without overrepresentation, yielding 1,754 cells across 26 cell types (mean 67 cells per type; range 25–250). After intersecting genes between the two datasets, 16,390 genes were available for downstream analysis.

### *2.2 Preprocessing*

Spatial and scRNA-seq count matrices were normalized identically using Scanpy v1.9.3 [14]: library size normalization to 10,000 counts followed by log1p transformation. Highly variable genes were selected with the Seurat v3 flavor [15], retaining the top 2,000 genes in each dataset. For each reference cell type, mean expression across cells was computed to form an initial 16,390 × 26 reference matrix.

### 2.3 Differentially Expressed Gene Selection

Cell-type-specific marker genes were selected using a Gini coefficient–based strategy following Erdogan and Eroglu [11]. For each gene *g*, the Gini coefficient across cell types is:

$$Gini(g) = (\Sigma_i \Sigma_j |x_i - x_j|) / (2n \Sigma_i x_i)$$

where $x_i$ is the mean expression of gene g in cell type i and n = 26. Values approach 0 for ubiquitously expressed genes and 1 for highly cell-type-specific genes. Three filters were applied: Gini ≥ 0.30; detection in at least 5% of cells in at least one cell type; and selection of the top 2,000 genes by Gini coefficient. This yielded a reference matrix X of dimension 2,000 × 26 with mean Gini = 0.857 (range 0.759–0.962).

### 2.4 WISpR Deconvolution

For each spot *j*, WISpR solves the regularized non-negative least squares problem

$$\beta^* = \text{argmin } ||y_j - X\beta||^2_2 + \lambda||\beta||^2_2 \text{ subject to } \beta \geq 0$$

where $\mathbf{y}_j$ is the spot's normalized expression vector restricted to the 2,000 selected genes, **X** is the reference matrix, and **β** is the vector of cell-type proportions to estimate. Sparsity is enforced via iterative soft-thresholding adapted from the SINDy framework [12]. Briefly, the algorithm initializes $\boldsymbol{\beta}^{(0)}$ by non-negative least squares, then iteratively updates

$$\beta_{temp} = \beta^{(k-1)} - (1/2\lambda) \cdot g, \quad \beta^{(k)} = max(\beta_{temp} - \tau, 0)$$

where g = −2XT(y − Xβ(k−1)) is the gradient. Iterations continue for up to 100 steps or until $||\beta(k) - \beta(k-1)||_2 < 10^{-6}$.

### 2.5 Hyperparameter Optimization

Spot-specific optimal (τ*, λ*) were selected via GridSearchCV (scikit-learn 1.3.0) with 5-fold cross-validation repeated 3 times. The search grid spanned τ ∈ [0.001, …, 0.500] and λ ∈ {0.001, 0.01, 0.1, 1.0, 10.0}. Cross-validation scored configurations by negative mean absolute error. Full optimization of all 2,110 spots required ~32 min on an Intel Core i5 workstation with 8 GB RAM.

### 2.6 Hyperparameter Sensitivity Analysis

To test whether spot-specific optimization is necessary, the GridSearch baseline was compared against four uniform, manually chosen configurations applied identically to every spot: loose threshold (τ = 0.05, λ = 10); strict threshold (τ = 0.30, λ = 10); weak penalty (τ = 0.15, λ = 5); and strong penalty (τ = 0.15, λ = 50). These configurations span the range of values a researcher might reasonably choose by inspection.

For each configuration, per-spot RMSE, $||y_j - X\beta_j||_2$, and sparsity (number of cell types with β > 0) were computed. Statistical comparison used the Wilcoxon signed-rank test on paired RMSE values across spots. Effect sizes were quantified with Cohen's d (parametric) and Cliff's delta (non-parametric), interpreted using standard thresholds.

### 2.7 Biological Validation

Predicted cell-type proportions were validated by correlation with canonical marker gene expression. Marker sets were defined for six brain cell categories: oligodendrocytes (*Mbp, Mog, Plp1, Olig1, Olig2*); neurons (*Slc17a7, Snap25, Rbfox3, Tubb3, Map2*); astrocytes (*Gfap, Aqp4, Aldh1l1, Slc1a2, Slc1a3*); immune cells (*Cx3cr1, Ptprc, Aif1, C1qa, C1qb*); vascular cells (*Cldn5, Pecam1, Flt1, Vwf, Tie1*); and ependymal cells (*Foxj1, Tmem212, Dynlrb2*). For each category, a composite marker score was computed as the mean expression of available markers per spot. Pearson and Spearman correlations between marker scores and predicted proportions were computed across all spots.

### 2.8 Implementation and Reproducibility

Analyses were performed in Python 3.13 (NumPy 1.24.3, SciPy 1.11.1, scikit-learn 1.3.0, Scanpy 1.9.3, pandas 2.0.3) and R 4.3.1 (Seurat 4.3.0). Visualizations used Matplotlib 3.7.2 and Seaborn 0.12.2. Random seeds were fixed (seed = 42) for all stochastic procedures.

## 3 RESULTS

### *3.1 GridSearch-Optimized WISpR Achieves Biologically Realistic Deconvolution*

Applied to 2,110 Visium spots with spot-specific hyperparameter optimization, WISpR produced a mean sparsity of 4.22 cell types per spot (median 4, SD 1.78), consistent with the expected cellular complexity of 55 µm Visium spots (Figure 1A). Approximately 74% of spots contained 3–6 cell types and only 3.2% contained more than seven, indicating that the algorithm successfully enforces biological sparsity. Reconstruction error was low overall: median RMSE = 0.280, with 75% of spots below 0.596 (Figure 1B). The RMSE distribution was right-skewed, with the largest errors concentrated at tissue boundaries (see Section 3.3).

The distributions of spot-specific optimal hyperparameters revealed consistent patterns. The threshold parameter $\tau$ was strongly skewed, with 82.7% of spots converging to $\tau \approx 0.50$ (the upper bound of the grid; Figure 1C), indicating that aggressive thresholding is required to achieve biological sparsity in most regions. The penalty parameter $\lambda$ was unimodal and centered near $\log10(\lambda) = 0$, i.e. $\lambda \approx 1.0$ (Figure 1D), suggesting moderate L2 regularization is generally optimal. Only 8.4% of spots required $\lambda > 10$, primarily in regions of high cellular diversity.

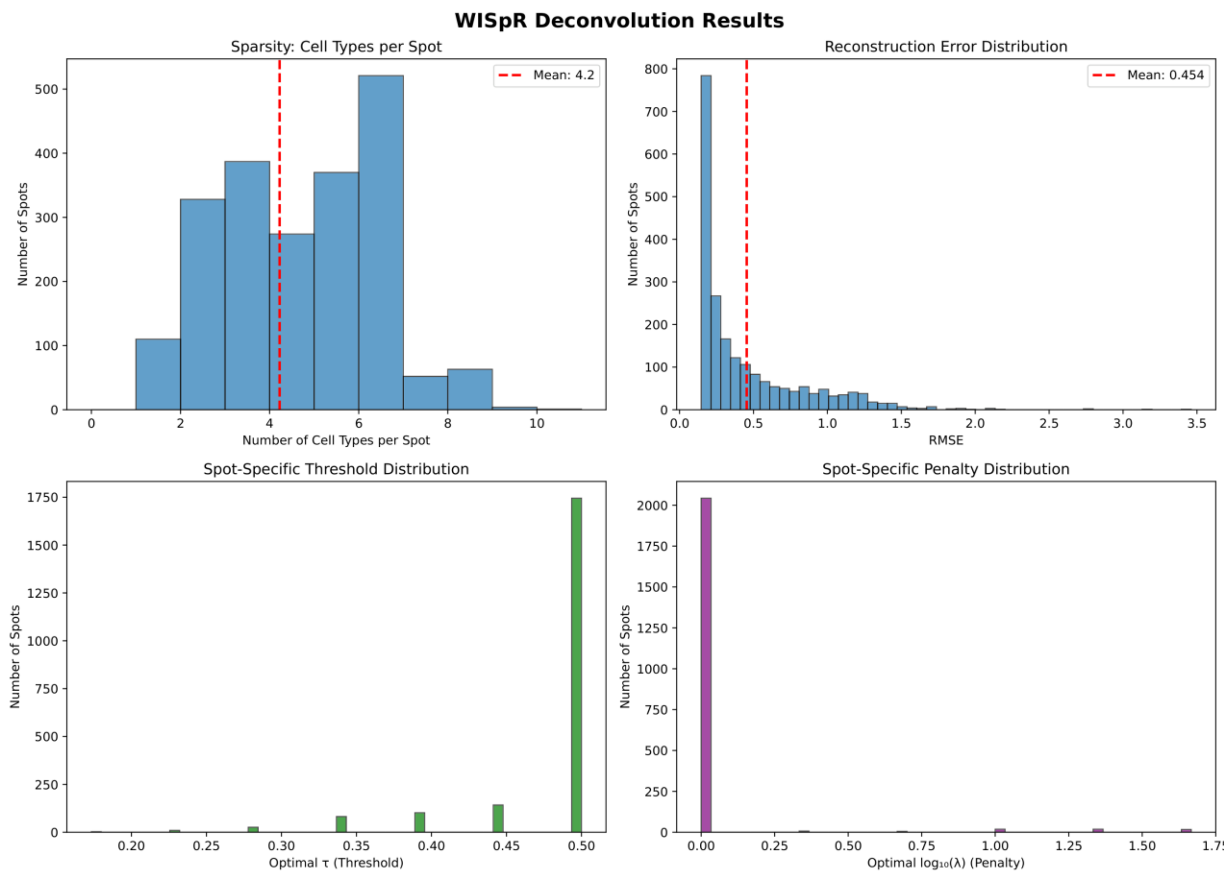


**Figure 1. WISpR deconvolution performance on mouse hippocampus. (A) Distribution of cell-type sparsity across 2,110 spots (mean 4.22 cell types per spot). (B) Distribution of reconstruction error (RMSE; mean 0.454, median 0.280). (C) Distribution of spot-specific optimal threshold parameter $\tau$; the majority of spots converge to the upper grid bound ($\tau \approx 0.50$). (D) Distribution of spot-specific optimal penalty parameter $\log10(\lambda)$; most spots prefer moderate regularization ($\lambda \approx 1.0$).**

### *3.2 Predictions Are Validated by Marker Gene Expression*

Predicted cell-type proportions correlated strongly with canonical marker gene expression across spots (Figure 2). Oligodendrocyte markers showed near-perfect correlation with two predicted clusters: Cluster_1_Oligos (r = 0.949, p

$< 10^{-300}$) and Cluster_10_Neurons ($r = 0.955$, $p < 10^{-300}$). Despite the "Neurons" name inherited from the Linnarsson atlas, Cluster_10 clearly represents an oligodendrocyte subpopulation. This illustrates a useful general property of marker-based validation: it can detect annotation errors in the reference that algorithmic deconvolution alone could not reveal.

Immune markers correlated strongly with Cluster_35_Immune ($r = 0.814$, $p < 10^{-250}$), consistent with microglia and infiltrating immune cells distributed across tissue. Three vascular clusters—Cluster_70_Vascular, Cluster_73_Blood, and Cluster_69_Vascular—exhibited consistent correlations with endothelial markers ($r = 0.79$–$0.80$, all $p < 10^{-230}$), likely reflecting distinct vascular subtypes (endothelial cells, pericytes, vascular smooth muscle) that share pan-vascular markers. Negative correlations between mutually exclusive cell types (e.g., oligodendrocytes vs. neurons) further support biological validity. Across all 156 marker–prediction comparisons, 23 reached statistical significance ($|r| > 0.3$, $p < 0.01$).

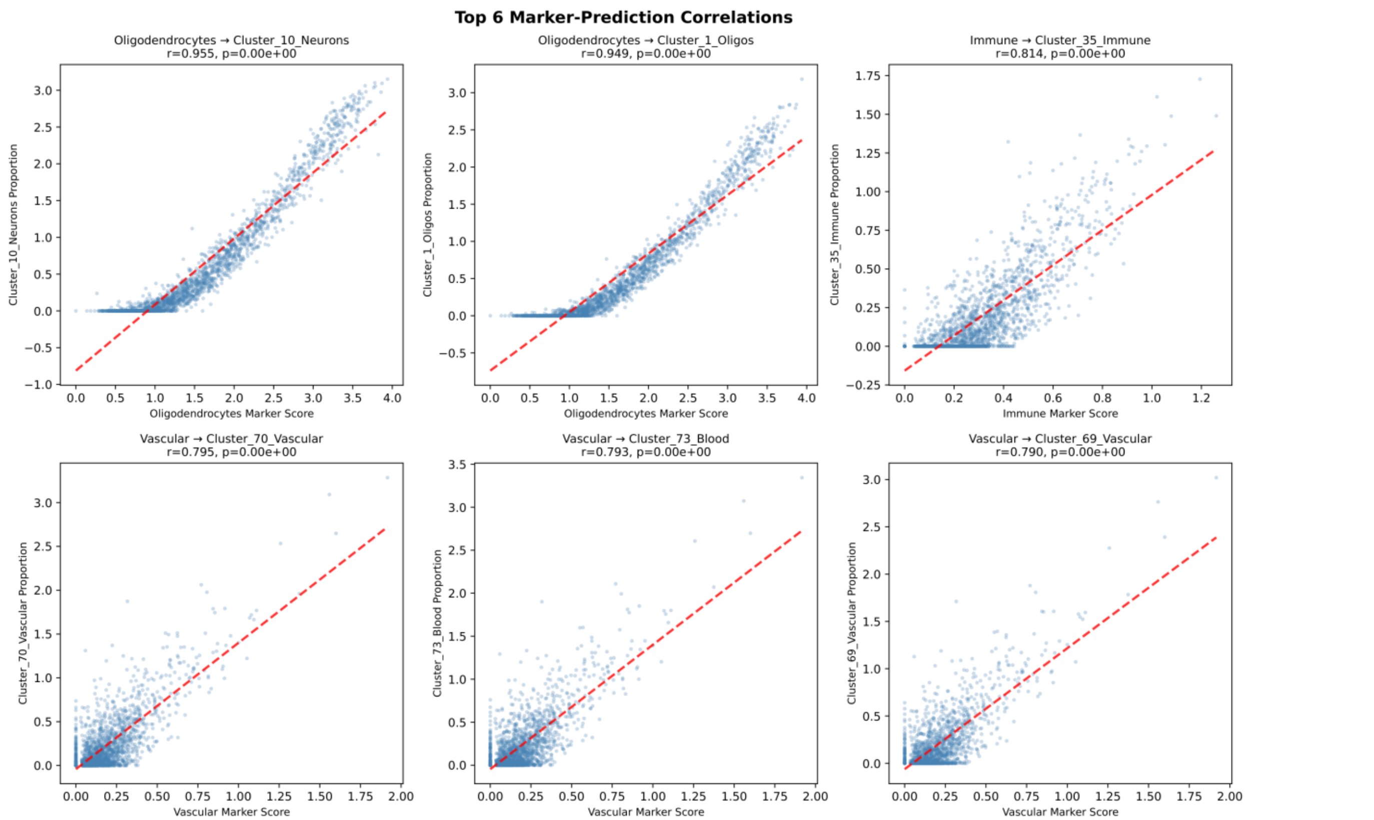


**Figure 2. Marker-gene validation of predicted cell-type proportions. Top six marker–prediction scatter plots, each point representing one spot (all $p < 10^{-230}$). Oligodendrocyte markers correlate strongly with Cluster_1_Oligos ($r = 0.949$) and with Cluster_10_Neurons ($r = 0.955$), the latter indicating mislabeling in the reference atlas. Immune markers correlate with Cluster_35_Immune ($r = 0.814$), and vascular markers with three vascular/blood clusters ($r = 0.79$–$0.80$). Red dashed lines indicate linear fits.**

### *3.3 Spatial Organization Recapitulates Hippocampal Anatomy*

Spatial mapping of predicted cell types revealed anatomically coherent patterns (Figure 3A). The six most prevalent cell types were detected in 54–73% of spots. Oligodendrocyte clusters localized to the central tissue region consistent with white matter enrichment, with high mean proportions (0.76–0.84) in dominant spots. Vascular clusters exhibited the expected scattered, network-like distribution. Immune cells showed diffuse coverage, consistent with surveillance microglia. Rare neuronal subtypes (Cluster_54/58/22_Neurons), detected in only 2.6–5.4% of spots, were restricted to specific anatomical subregions (Supplementary Figure S3), consistent with layer-specific or functionally specialized populations.

Dominant cell-type maps (Figure 3B) showed Cluster_10_Neurons occupying the central region, vascular cells scattered throughout, and Cluster_1_Oligos forming distinct patches. Confidence (proportion of the dominant type) was highest in the tissue core and decreased toward edges. Sparsity maps (Figure 3C) showed central regions with 3–5 cell types and elevated complexity (6–10 types) at boundaries. Reconstruction error followed an inverse pattern: lowest at the core and elevated at edges and the lower-left region of the tissue. These edge effects likely reflect partial spots and reduced RNA capture quality at section boundaries rather than algorithmic failure, a known limitation of Visium platforms.

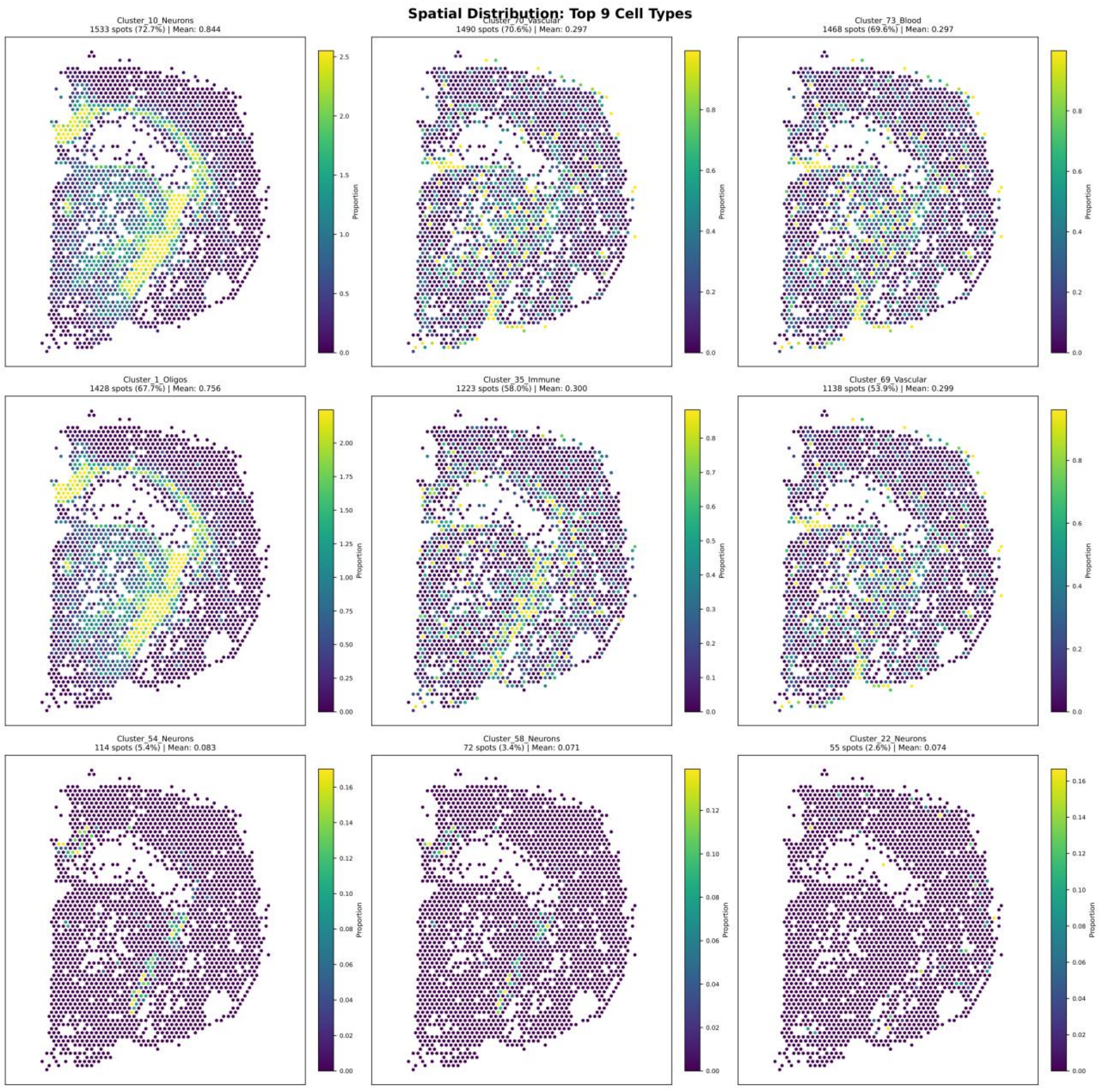


**Figure 3. Spatial organization of predicted cell types. Spatial heatmaps of the nine most prevalent predicted cell types across the hippocampus section; color indicates predicted proportion per spot. Oligodendrocyte clusters (Cluster_1_Oligos, Cluster_10_Neurons) localize to central white-matter regions, vascular clusters show a scattered network-like distribution, and rare neuronal subtypes (Cluster_54/58/22_Neurons; 2.6–5.4% of spots) are restricted to specific anatomical subregions.**

### *3.4 Manual Hyperparameter Selection Performs Substantially Worse*

To test whether the computational cost of GridSearch is justified, the baseline was compared against four uniform manual configurations (Figure 4, Table 1). All four performed substantially worse, both statistically and biologically.

**Table 1.** Performance comparison of hyperparameter configurations.

| Configuration | τ | λ | Mean RMSE | Sparsity | % Worse | Cohen's d | Cliff's δ |
|---|---|---|---|---|---|---|---|
| Baseline (GridSearch) | var. | var. | 0.454 | 4.22 | — | — | — |
| Loose threshold | 0.05 | 10 | 5.243 | 25.99 | +91.3% | −2.64 | −0.999 |
| Strict threshold | 0.30 | 10 | 4.840 | 20.02 | +90.6% | −2.41 | −0.997 |
| Weak penalty | 0.15 | 5 | 10.505 | 25.97 | +95.7% | −2.76 | −1.000 |
| Strong penalty | 0.15 | 50 | 0.779 | 12.58 | +41.6% | −0.77 | −0.529 |

*All Wilcoxon signed-rank p-values $< 10^{-100}$. Effect sizes: $|d| > 0.8$ = large; $|\delta| > 0.474$ = large. n = 2,110 spots per condition.*

The loose threshold configuration ($\tau = 0.05$, $\lambda = 10$) produced mean RMSE = 5.24, a 91.3% increase over baseline (Wilcoxon $p < 10^{-100}$, Cohen's $d = -2.64$). It also assigned a mean of 25.99 cell types per spot—essentially all 26 reference types—rendering it biologically meaningless. The strict threshold configuration ($\tau = 0.30$, $\lambda = 10$) failed similarly: RMSE = 4.84 (+90.6%, $p < 10^{-100}$, $d = -2.41$), sparsity = 20.0 cell types per spot. Even aggressive uniform thresholding cannot enforce sensible sparsity without adapting to local tissue context.

The weak penalty configuration ($\tau = 0.15$, $\lambda = 5$) was the worst overall, with RMSE = 10.51 (+95.7%, $p < 10^{-100}$, $d = -2.76$) and sparsity = 25.97 cell types per spot, indicating that insufficient regularization destabilizes the optimization and overfits noise. The strong penalty configuration ($\tau = 0.15$, $\lambda = 50$) was the best of the four manual settings, achieving RMSE = 0.78, but still 41.6% worse than baseline ($p < 10^{-100}$, $d = -0.77$) and predicting 12.58 cell types per spot—nearly three times the baseline.

Effect sizes for all four comparisons exceeded standard thresholds for large effects ($|d| > 0.8$), with Cliff's delta values of −0.997 to −1.000 for the three worst configurations, indicating that the GridSearch baseline outperforms manual settings on essentially every individual spot. Cumulative distribution functions (Figure 4) confirm that baseline superiority holds across the full RMSE distribution, not just in means.

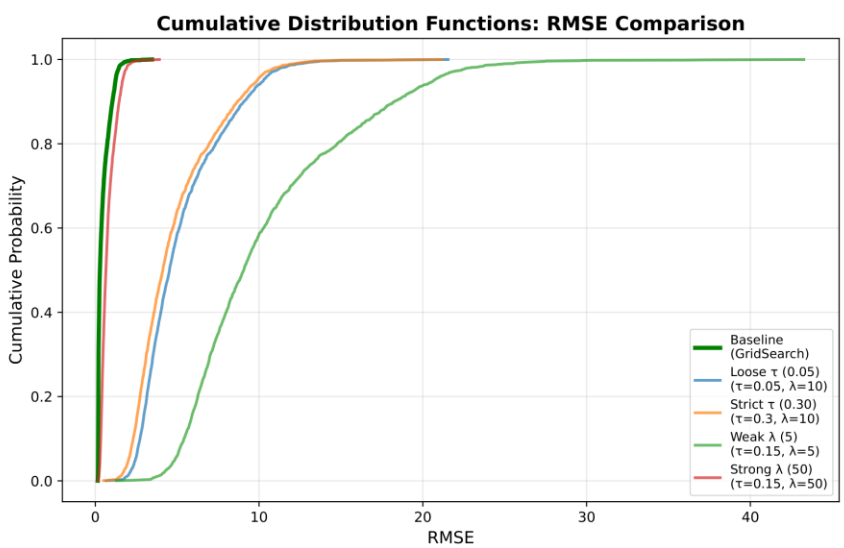


**Figure 4. GridSearch versus manual hyperparameter configurations. Cumulative distribution functions (CDFs) of per-spot RMSE for the GridSearch baseline (green) and the four manual configurations. The baseline CDF rises steeply at low RMSE, indicating consistently low reconstruction error, whereas all manual configurations are shifted toward substantially higher RMSE. The weak-penalty configuration (λ = 5) shows the heaviest tail. n = 2,110 spots per condition.**

## 4 DISCUSSION

The results establish that data-driven, spot-specific hyperparameter optimization is essential—not merely beneficial—for accurate WISpR deconvolution. All four uniform manual configurations failed substantially, with 41.6%–95.7% higher reconstruction error and 3–6× more cell types per spot than biologically plausible. Effect sizes well above conventional "large effect" thresholds (Cohen's d up to −2.76; Cliff's delta near −1.0) indicate that this is not a marginal performance difference but a categorical one: the baseline outperforms manual selection on essentially every spot.

The failure of manual selection reflects a basic mismatch between uniform hyperparameters and spatially heterogeneous tissue biology. Different regions—white matter, layered grey matter, vasculature, edges, transitional zones—have different cellular compositions, signal-to-noise characteristics, and gene expression variability. A single ($\tau$, $\lambda$) cannot accommodate all of these simultaneously. GridSearch addresses this by tuning each spot to its local context. That 82.7% of spots converge to the upper bound of the $\tau$ grid is itself informative: it suggests aggressive sparsity enforcement is generally optimal for Visium data, and that future implementations might profitably explore $\tau > 0.50$—though with care, since overly large thresholds risk excluding biologically meaningful rare cell types.

The biological validation results provide independent support for the GridSearch predictions. Correlations of $r \approx 0.95$ between predicted oligodendrocyte proportions and canonical oligodendrocyte markers, $r \approx 0.81$ for immune cells, and $r \approx 0.80$ for vascular cells across thousands of spots are unlikely to arise from chance or overfitting. Notably, Cluster_10_Neurons—labeled as neuronal in the Linnarsson reference—correlated strongly with oligodendrocyte markers, illustrating that reference atlas annotations are themselves hypotheses subject to validation. Marker-based checks should be a routine component of any deconvolution workflow.

Edge effects in the data warrant interpretation. Elevated RMSE and sparsity at tissue boundaries (Figure 3C) likely reflect technical artifacts: partial spots capturing incomplete cellular content, less uniform permeabilization at edges, and genuine cellular heterogeneity at anatomical interfaces. Importantly, marker correlations remained strong even when including edge spots, suggesting that predictions there are still biologically meaningful, just noisier. Edge artifacts are not unique to WISpR; they affect all spatial deconvolution methods [7, 10]. Future work could address them with explicit partial-spot models, spatial smoothing priors, or post-hoc filtering of low-confidence boundaries.

This study has several limitations. A single tissue (mouse hippocampus) with a matched reference was evaluated; cross-tissue, cross-species, or mismatched-reference scenarios remain to be tested. The hyperparameter grid, while extensive, was not exhaustive; the convergence of most spots to the upper $\tau$ bound suggests that larger ranges may further improve performance. WISpR was also not compared against other deconvolution methods (DWLS, RCTD, SPOTlight, Stereoscope, Cell2location); the scope here was specifically the necessity of hyperparameter optimization within WISpR. Finally, the computational cost of GridSearch (~32 min for 2,110 spots) may become limiting for higher-resolution platforms such as Visium-HD, motivating future development of accelerated tuning strategies—surrogate modeling, Bayesian optimization, or cluster-level optimization that groups similar spots.

In summary, GridSearch-based, spot-specific hyperparameter optimization has been shown to be essential for WISpR to produce accurate, biologically meaningful spatial deconvolutions. Uniform manual hyperparameters—regardless of how reasonably they are chosen—cannot capture the spatial heterogeneity of real tissues, and their use produces both elevated reconstruction error and biologically implausible sparsity. These findings establish data-driven optimization as a default expectation for WISpR application, and highlight a methodological principle likely applicable to spatial deconvolution more broadly.

## DATA AND CODE AVAILABILITY

Mouse hippocampus Visium data are publicly available from 10X Genomics. The Linnarsson scRNA-seq reference (l1_hippocampus.loom) is available from the Linnarsson Laboratory Mouse Brain Atlas. All analysis code, intermediate files, and figures generated in this study will be deposited at the GitHub repository upon manuscript acceptance. Additional details of the underlying WISpR method are available in the original publication – the main reference paper – by Erdogan and Eroglu [11].

## REFERENCES


**[1]** Ståhl PL, Salmén F, Vickovic S, et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. Science. 2016;353(6294):78–82.

**[2]** Rodriques SG, Stickels RR, Goeva A, et al. Slide-seq: A scalable technology for measuring genome-wide expression at high spatial resolution. Science. 2019;363(6434):1463–1467.

**[3]** Asp M, Bergenstråhle J, Lundeberg J. Spatially resolved transcriptomes—next generation tools for tissue exploration. BioEssays. 2020;42(10):1900221.

**[4]** Moses L, Pachter L. Museum of spatial transcriptomics. Nat Methods. 2022;19(5):534–546.

**[5]** Longo SK, Guo MG, Ji AL, Khavari PA. Integrating single-cell and spatial transcriptomics to elucidate intercellular tissue dynamics. Nat Rev Genet. 2021;22(10):627–644.

**[6]** Tsoucas D, Dong R, Chen H, Zhu Q, Guo G, Yuan GC. Accurate estimation of cell-type composition from gene expression data. Nat Commun. 2019;10(1):2975.

**[7]** Cable DM, Murray E, Zou LS, et al. Robust decomposition of cell type mixtures in spatial transcriptomics. Nat Biotechnol. 2022;40(4):517–526.

**[8]** Elosua-Bayes M, Nieto P, Mereu E, Gut I, Heyn H. SPOTlight: seeded NMF regression to deconvolute spatial transcriptomics spots with single-cell transcriptomes. Nucleic Acids Res. 2021;49(9):e50.

**[9]** Andersson A, Bergenstråhle J, Asp M, et al. Single-cell and spatial transcriptomics enables probabilistic inference of cell type topography. Commun Biol. 2020;3(1):565.

**[10]** Kleshchevnikov V, Shmatko A, Dann E, et al. Cell2location maps fine-grained cell types in spatial transcriptomics. Nat Biotechnol. 2022;40(5):661–671.

**[11]** Erdogan E, Eroglu A. Sparse deconvolution of cell-type medleys in spatial transcriptomics. Bioinform Adv. 2025;5(1):vbae190.

**[12]** Brunton SL, Proctor JL, Kutz JN. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. Proc Natl Acad Sci USA. 2016;113(15):3932–3937.

**[13]** Zeisel A, Hochgerner H, Lönnerberg P, et al. Molecular architecture of the mouse nervous system. Cell. 2018;174(4):999–1014.

**[14]** Wolf FA, Angerer P, Theis FJ. SCANPY: large-scale single-cell gene expression data analysis. Genome Biol. 2018;19(1):15.

**[15]** Stuart T, Butler A, Hoffman P, et al. Comprehensive integration of single-cell data. Cell. 2019;177(7):1888–1902.

**[16]** Dong R, Yuan GC. SpatialDWLS: accurate deconvolution of spatial transcriptomic data. Genome Biol. 2021;22(1):145.

**[17]** Ma Y, Zhou X. Spatially informed cell-type deconvolution for spatial transcriptomics. Nat Biotechnol. 2022;40(9):1349–1359.